

DECISION TREE REGRESSION Assignment

R-Score with different parameter combinations, for every R-Score, we are creating a new model

S.No	Criterion	Max Features	Splitter	RValue
1	Absolute_error	auto	best	0.9493726735323613
2	Absolute_error	auto	random	0.5780023619811279
3	Absolute_error	sqrt	best	0.5642127299861528
4	Absolute_error	sqrt	random	-0.5014681210604035
5	Absolute_error	Log2	best	0.5752657964430266
6	Absolute_error	Log2	random	0.2174827794013896
7	poisson	auto	best	0.9318880262356684
8	poisson	auto	random	0.8876629502127538
9	poisson	sqrt	best	0.42313512471580794
10	poisson	sqrt	random	0.7524659638413829
11	poisson	Log2	best	0.8882213599783344
12	poisson	Log2	random	0.9454810225284668
13	Friedman_mse	auto	best	0.9141972921025665
14	Friedman_mse	auto	random	0.9324051807868639
15	Friedman_mse	sqrt	best	0.5639936898293063
16	Friedman_mse	sqrt	random	0.610034721887075
17	Friedman_mse	Log2	best	0.6586406993603734
18	Friedman_mse	Log2	random	0.25501938600153373

Absolute_error

auto

best

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```
[22]: #Model Creation code for DecisionTree
[23]: from sklearn.tree import DecisionTreeRegressor
[39]: regressor = DecisionTreeRegressor(criterion='absolute_error', splitter='best')
[40]: regressor = regressor.fit(x_train, y_train)
[41]: #Code to see how the tree is plotted
[42]: import matplotlib.pyplot as plt
[43]: from sklearn import tree
[44]: tree.plot_tree(regressor)
[44]: [Text(0.46875, 0.9444444444444444, 'x[0] <= 64036.785\nabsolute_error = 34660.428\nsamples = 35\nvalue = 108733.99'),
Text(0.203125, 0.8333333333333334, 'x[0] <= 8410.595\nabsolute_error = 19034.017\nsamples = 14\nvalue = 70628.735'),
Text(0.3359375, 0.8888888888888888, 'True'),
Text(0.10416666666666667, 0.7222222222222222, 'x[0] <= 771.14\nabsolute_error = 12812.404\nsamples = 5\nvalue = 42559.73'),
Text(0.0625, 0.6111111111111112, 'x[2] <= 22586.529\nabsolute_error = 9292.777\nsamples = 3\nvalue = 35673.41'),
Text(0.04166666666666667, 0.5, 'x[1] <= 93585.035\nabsolute_error = 3443.16\nsamples = 2\nvalue = 39116.57'),
Text(0.020833333333333332, 0.3888888888888889, 'absolute_error = 0.0\nsamples = 1\nvalue = 35673.41'),
Text(0.0625, 0.3888888888888889, 'absolute_error = 0.0\nsamples = 1\nvalue = 42559.73'),
Text(0.08333333333333333, 0.5, 'absolute_error = 0.0\nsamples = 1\nvalue = 14681.4'),
Text(0.14583333333333334, 0.6111111111111112, 'x[0] <= 1157.845\nabsolute_error = 7717.665\nsamples = 2\nvalue = 57208.415'),
Text(0.125, 0.5, 'absolute_error = 0.0\nsamples = 1\nvalue = 64928.08'),
Text(0.16666666666666666, 0.5, 'absolute_error = 0.0\nsamples = 1\nvalue = 49490.75'),
Text(0.3020833333333333, 0.7222222222222222, 'x[0] <= 26152.345\nabsolute_error = 10369.45\nsamples = 9\nvalue = 81005.76'),
Text(0.22916666666666666, 0.6111111111111112, 'x[2] <= 31934.446\nabsolute_error = 2099.387\nsamples = 3\nvalue = 69758.98'),
Text(0.20833333333333334, 0.5, 'absolute_error = 0.0\nsamples = 1\nvalue = 65200.33'),
Text(0.25, 0.5, 'x[2] <= 91767.641\nabsolute_error = 869.755\nsamples = 2\nvalue = 70628.735'),
Text(0.22916666666666666, 0.3888888888888889, 'absolute_error = 0.0\nsamples = 1\nvalue = 49758.98'),
Text(0.2708333333333333, 0.3888888888888889, 'absolute_error = 0.0\nsamples = 1\nvalue = 71498.49'),
Text(0.375, 0.6111111111111112, 'x[0] <= 41314.23\nabsolute_error = 6479.802\nsamples = 6\nvalue = 90328.665'),
Text(0.3333333333333333, 0.5, 'x[1] <= 122801.129\nabsolute_error = 4156.093\nsamples = 3\nvalue = 81005.76'),
Text(0.3125, 0.3888888888888889, 'x[0] <= 33656.421\nabsolute_error = 1382.925\nsamples = 2\nvalue = 79622.835')]
```

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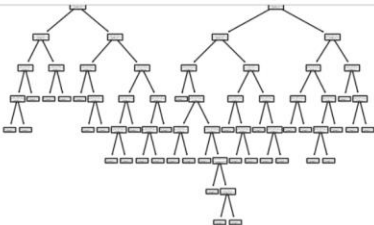
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JupyterLab Python [conda envbase] Anaconda Toolbox



```
[45]: plt.show()
[46]: y_pred = regressor.predict(x_test)
[47]: #Evaluation Metrics
[48]: from sklearn.metrics import r2_score
[49]: r_score = r2_score(y_test, y_pred)
[51]: r_score
[51]: 0.949372673532361
```

Absolute_error

auto

random

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JupyterLab Python [conda envbase] Trusted

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```
[22]: #Model Creation code for DecisionTree
[23]: from sklearn.tree import DecisionTreeRegressor
[52]: regressor = DecisionTreeRegressor(criterion='absolute_error', splitter='random')
[53]: regressor = regressor.fit(x_train, y_train)
[54]: #Code to see how the tree is plotted
[55]: import matplotlib.pyplot as plt
[56]: from sklearn import tree
[57]: tree.plot_tree(regressor)

[Text(0.5556891025641025, 0.95, 'x[2] <= 310904.089\nabsolute_error = 34660.428\nsamples = 35\nvalue = 108733.99'),
Text(0.23958333333333334, 0.85, 'x[0] <= 200.797\nabsolute_error = 28374.311\nsamples = 29\nvalue = 101004.64'),
Text(0.39763621794871795, 0.8999999999999999, 'True '),
Text(0.07692307692307693, 0.75, 'x[2] <= 38845.145\nabsolute_error = 13939.165\nsamples = 2\nvalue = 28620.565'),
Text(0.05128205128205128, 0.65, 'absolute_error = 0.0\nsamples = 1\nvalue = 42559.73'),
Text(0.10256410256410256, 0.65, 'absolute_error = 0.0\nsamples = 1\nvalue = 14681.4'),
Text(0.40224358974358976, 0.75, 'x[0] <= 55088.768\nabsolute_error = 24877.302\nsamples = 27\nvalue = 107404.34'),
Text(0.15384615384615385, 0.65, 'x[0] <= 25212.487\nabsolute_error = 12635.194\nsamples = 10\nvalue = 70628.735'),
Text(0.10256410256410256, 0.55, 'x[2] <= 181988.019\nabsolute_error = 9394.593\nsamples = 6\nvalue = 65863.205'),
Text(0.07692307692307693, 0.45, 'x[1] <= 63267.507\nabsolute_error = 8131.596\nsamples = 5\nvalue = 65200.33'),
Text(0.05128205128205128, 0.35, 'absolute_error = 0.0\nsamples = 1\nvalue = 35673.41'),
Text(0.10256410256410256, 0.35, 'x[2] <= 68315.295\nabsolute_error = 2782.765\nsamples = 4\nvalue = 67479.655'),
Text(0.07692307692307693, 0.25, 'x[2] <= 32959.129\nabsolute_error = 1618.967\nsamples = 3\nvalue = 65200.33'),
Text(0.05128205128205128, 0.15, 'x[1] <= 136815.6\nabsolute_error = 137.125\nsamples = 2\nvalue = 65863.205'),
Text(0.02564102564102564, 0.05, 'absolute_error = 0.0\nsamples = 1\nvalue = 64926.00'),
Text(0.07692307692307693, 0.05, 'absolute_error = 0.0\nsamples = 1\nvalue = 65200.33')]
```

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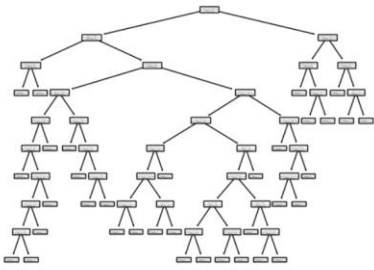
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```
[58]: plt.show()
[59]: y_pred = regressor.predict(x_test)
[60]: #Evaluation Metrics
[61]: from sklearn.metrics import r2_score
[62]: r_score = r2_score(y_test, y_pred)

[63]: r_score
[63]: 0.5788023619811279
```

Absolute_error, random, max_features=sqrt

Absolute_error	sqrt	random
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JupyterLab Python [conda env:base] Anaconda Toolbox

```
[22]: #Model Creation code for DecisionTree
[23]: from sklearn.tree import DecisionTreeRegressor
[64]: regressor = DecisionTreeRegressor(criterion='absolute_error', splitter='random', max_features='sqrt')
[65]: regressor = regressor.fit(x_train, y_train)
[66]: #Code to see how the tree is plotted
[67]: import matplotlib.pyplot as plt
[68]: from sklearn import tree
[69]: tree.plot_tree(regressor)
[69]: [Text(0.3950892857142857, 0.95, 'x[2] <= 88828.866\nabsolute_error = 34660.428\nsamples = 35\nvalue = 108733.99'),
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Text(0.07142857142857142, 0.65, 'x[0] <= 777.974\nabsolute_error = 9750.89\nsamples = 3\nvalue = 42559.73'),
Text(0.047619047619047616, 0.55, 'x[0] <= 393.239\nabsolute_error = 3443.16\nsamples = 2\nvalue = 39116.57'),
Text(0.023809523809523808, 0.45, 'absolute_error = 0.0\nsamples = 1\nvalue = 42559.73'),
Text(0.07142857142857142, 0.45, 'absolute_error = 0.0\nsamples = 1\nvalue = 35673.41'),
Text(0.09523809523809523, 0.55, 'absolute_error = 0.0\nsamples = 1\nvalue = 64926.08'),
Text(0.16666666666666666, 0.65, 'x[2] <= 32671.073\nabsolute_error = 21826.272\nsamples = 4\nvalue = 67479.655'),
Text(0.14285714285714285, 0.55, 'absolute_error = 0.0\nsamples = 1\nvalue = 65200.33'),
Text(0.19047619047619047, 0.55, 'x[2] <= 37522.262\nabsolute_error = 27582.147\nsamples = 3\nvalue = 69758.98'),
Text(0.16666666666666666, 0.45, 'absolute_error = 0.0\nsamples = 1\nvalue = 69758.98'),
Text(0.21428571428571427, 0.45, 'x[0] <= 15243.327\nabsolute_error = 41373.22\nsamples = 2\nvalue = 56054.62'),
Text(0.19047619047619047, 0.35, 'absolute_error = 0.0\nsamples = 1\nvalue = 14681.4'),
Text(0.23809523809523808, 0.35, 'absolute_error = 0.0\nsamples = 1\nvalue = 97427.84'),
Text(0.16666666666666666, 0.75, 'absolute_error = 0.0\nsamples = 1\nvalue = 122776.86'),
Text(0.6473214285714286, 0.85, 'x[2] <= 243673.084\nabsolute_error = 29143.979\nsamples = 27\nvalue = 125370.37'),
Text(0.5212053571428572, 0.8999999999999999, 'False'),
Text(0.48214285714285715, 0.75, 'x[4] <= 0.103\nabsolute_error = 16161.963\nsamples = 10\nvalue = 95322.89'),
Text(0.4166666666666667, 0.65, 'x[3] <= 0.493\nabsolute_error = 16684.916\nsamples = 8\nvalue = 90328.665')]
```

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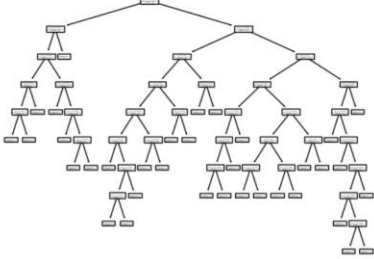
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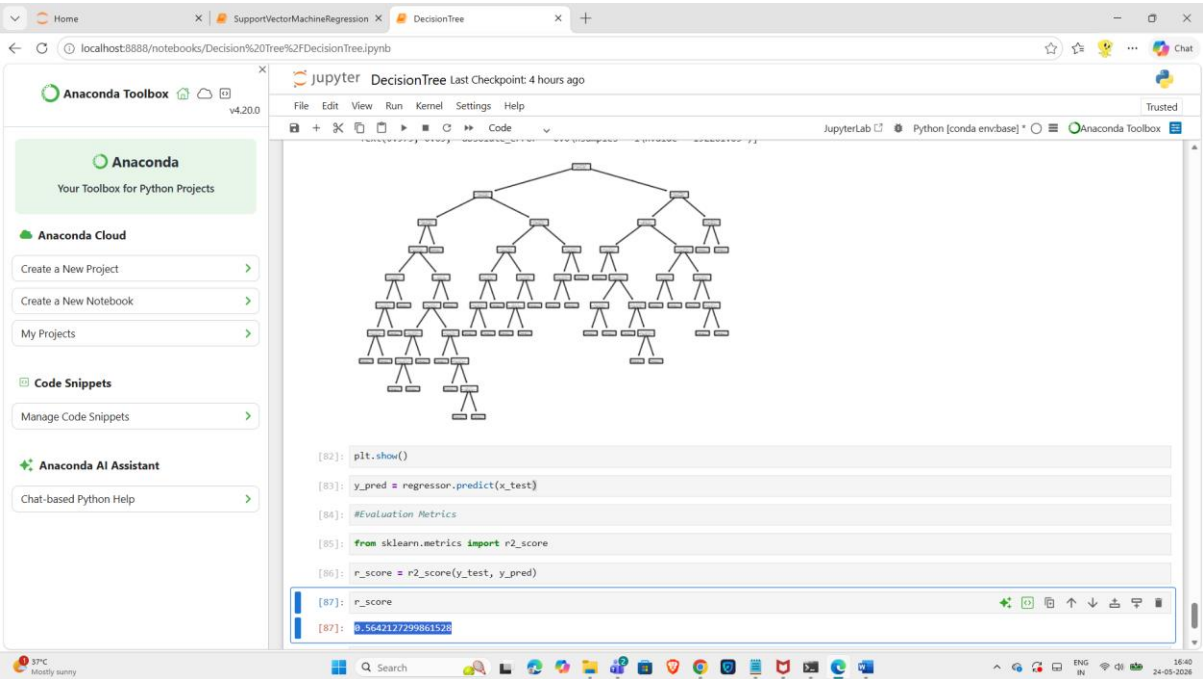
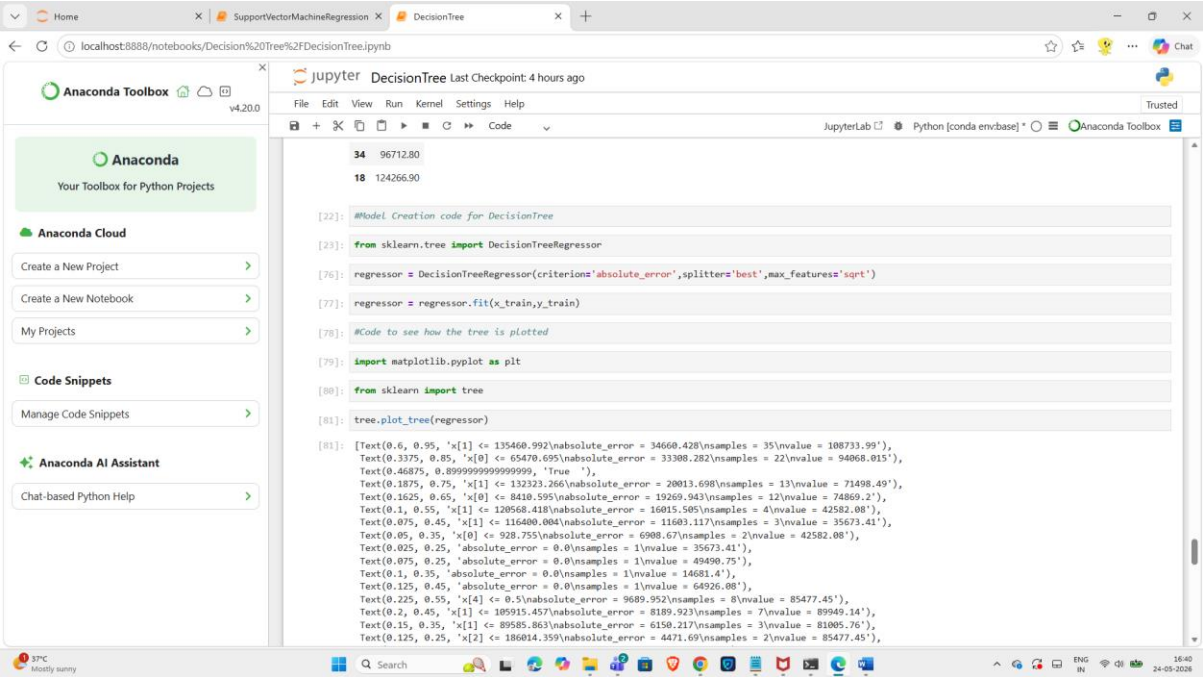
```
Text(0.9761904761904762, 0.25, 'absolute_error = 0.0\nsamples = 1\nvalue = 192261.83'),
Text(0.9523809523809523, 0.55, 'absolute_error = 0.0\nsamples = 1\nvalue = 111313.02')]
```



```
[70]: plt.show()
[71]: y_pred = regressor.predict(x_test)
[72]: #Evaluation Metrics
[73]: from sklearn.metrics import r2_score
[74]: r_score = r2_score(y_test, y_pred)
[75]: r_score
[75]: 0.9016651210604032
```

Absolute_error, best, max_features=sqrt

Absolute_error	sqrt	best
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Absolute_error

Log2

best

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34 96712.80
18 124266.90

```
[22]: #Model Creation code for DecisionTree
[23]: from sklearn.tree import DecisionTreeRegressor
[88]: regressor = DecisionTreeRegressor(criterion='absolute_error', splitter='best', max_features='log2')
[89]: regressor = regressor.fit(x_train, y_train)
[90]: #Code to see how the tree is plotted
[91]: import matplotlib.pyplot as plt
[92]: from sklearn import tree
[93]: tree.plot_tree(regressor)
[93]: [Text(0.6183723404255319, 0.9375, 'x[1] <= 135460.992\nabsolute_error = 34660.428\nsamples = 35\nvalue = 108733.99'),
Text(0.34308510638297873, 0.8125, 'x[0] <= 65470.695\nabsolute_error = 33308.282\nsamples = 22\nvalue = 94068.015'),
Text(0.4767287234042553, 0.875, 'True '),
Text(0.16489361702127658, 0.6875, 'x[0] <= 8410.595\nabsolute_error = 20013.698\nsamples = 13\nvalue = 71498.49'),
Text(0.0425531914893617, 0.5625, 'x[1] <= 83779.68\nabsolute_error = 12812.408\nsamples = 5\nvalue = 42559.73'),
Text(0.02127659574468085, 0.4375, 'absolute_error = 0.0\nsamples = 1\nvalue = 35673.41'),
Text(0.06382978723404255, 0.4375, 'x[1] <= 116400.004\nabsolute_error = 14293.925\nsamples = 4\nvalue = 46025.24'),
Text(0.0425531914893617, 0.3125, 'absolute_error = 0.0\nsamples = 1\nvalue = 49490.75'),
Text(0.0851063829787234, 0.3125, 'x[1] <= 120568.418\nabsolute_error = 16748.227\nsamples = 3\nvalue = 42559.73'),
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Text(0.1702127659574468, 0.3125, 'x[0] <= 19573.33\nabsolute_error = 869.755\nsamples = 2\nvalue = 70628.735')]
```

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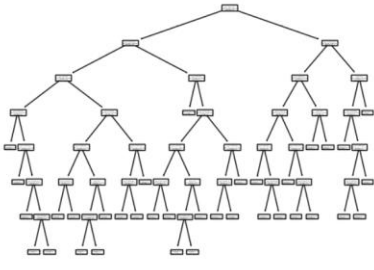
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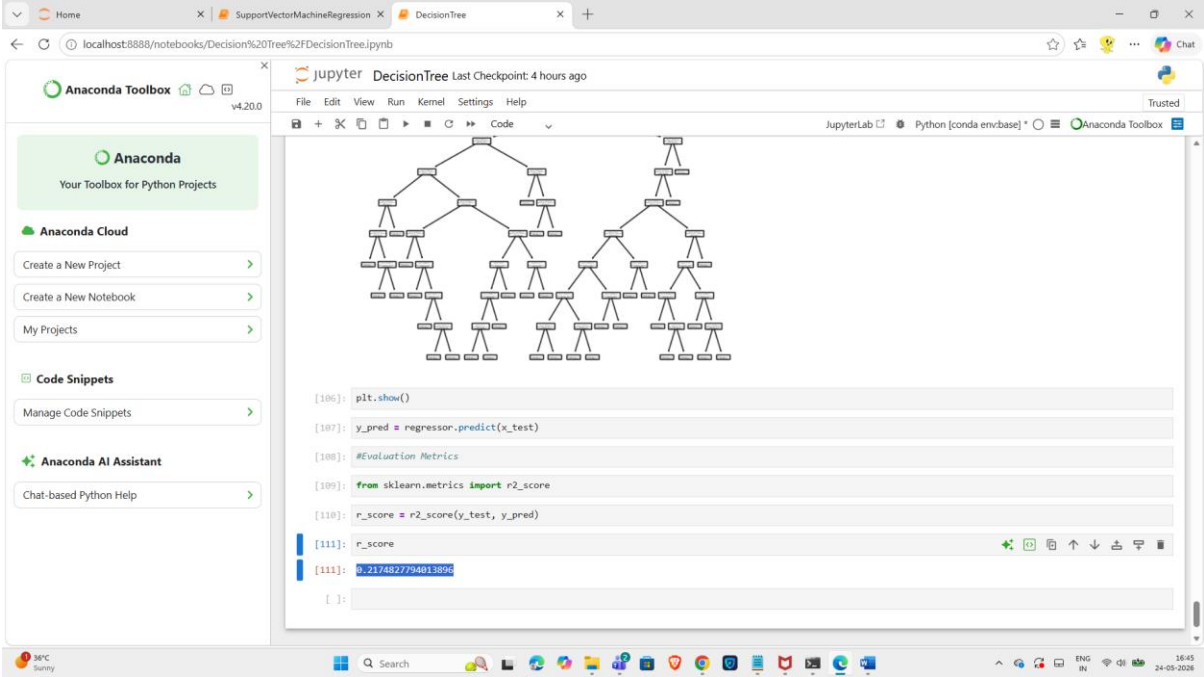
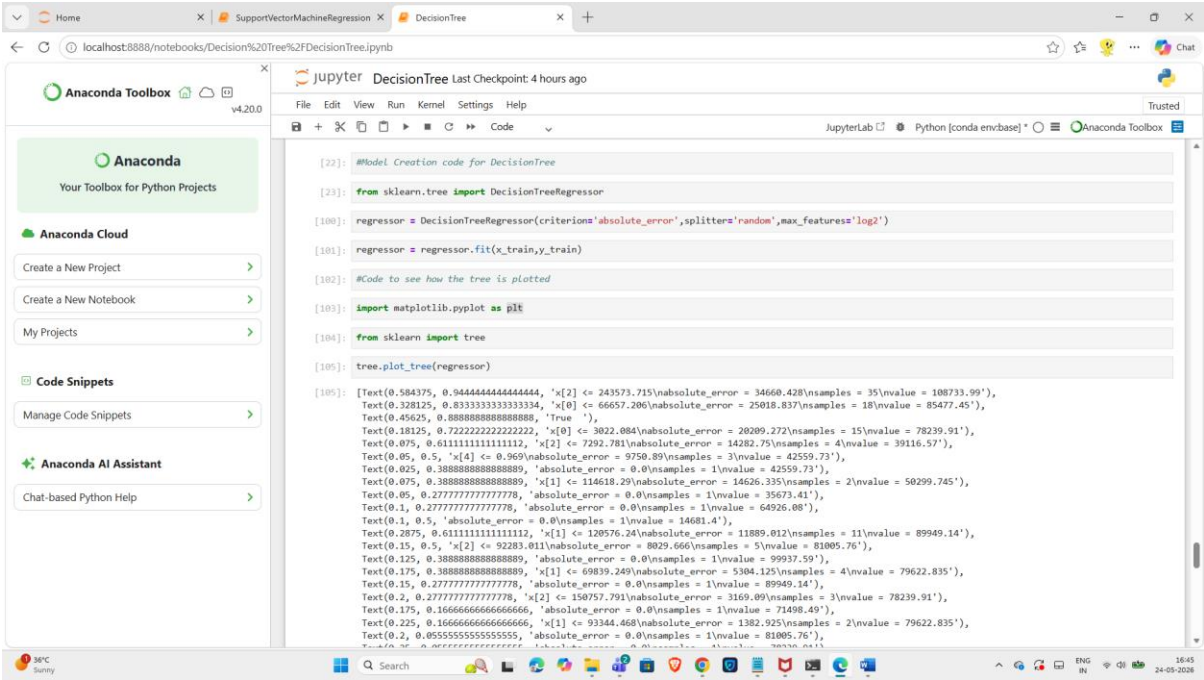
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Text(0.9787234042553191, 0.5625, 'absolute_error = 0.0\nsamples = 1\nvalue = 132602.65')]



```
[94]: plt.show()
[95]: y_pred = regressor.predict(x_test)
[96]: #Evaluation Metrics
[97]: from sklearn.metrics import r2_score
[98]: r_score = r2_score(y_test, y_pred)
[99]: r_score
[99]: 0.5752657964430266
```


Absolute_error	Log2	random
----------------	------	--------



poisson

auto

best

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```
[22]: #Model Creation code for DecisionTree
[23]: from sklearn.tree import DecisionTreeRegressor

[117]: regressor = DecisionTreeRegressor(criterion='poisson', splitter='best', max_features=None)
[118]: regressor = regressor.fit(x_train, y_train)
[119]: #Code to see how the tree is plotted
[120]: import matplotlib.pyplot as plt
[121]: from sklearn import tree
[122]: tree.plot_tree(regressor)

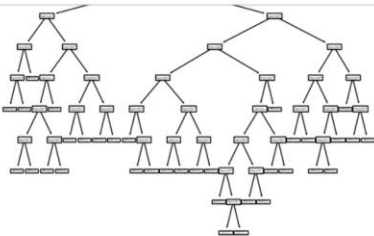
[123]: [Text(0.425, 0.9444444444444444, 'x[0] <= 53032.215\npoisson = 9248.617\nsamples = 35\nvalue = 110225.321'),
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Text(0.06, 0.7222222222222222, 'x[2] <= 22586.529\npoisson = 2536.582\nsamples = 3\nvalue = 30971.513'),
Text(0.04, 0.6111111111111112, 'x[1] <= 93585.035\npoisson = 151.735\nsamples = 2\nvalue = 39116.57'),
Text(0.02, 0.5, 'poisson = 0.0\nsamples = 1\nvalue = 35673.41'),
Text(0.06, 0.5, 'poisson = -0.0\nsamples = 1\nvalue = 42559.73'),
Text(0.08, 0.6111111111111112, 'poisson = 0.0\nsamples = 1\nvalue = 14681.4'),
Text(0.18, 0.7222222222222222, 'x[0] <= 26152.345\npoisson = 1875.385\nsamples = 9\nvalue = 73419.737'),
Text(0.12, 0.6111111111111112, 'x[3] <= 0.5\npoisson = 495.34\nsamples = 5\nvalue = 64174.926'),
Text(0.1, 0.5, 'x[2] <= 31934.446\npoisson = 59.835\nsamples = 4\nvalue = 67845.97'),
Text(0.06, 0.3888888888888889, 'x[2] <= 15119.325\npoisson = 0.145\nsamples = 2\nvalue = 65063.205'),
Text(0.04, 0.2777777777777778, 'poisson = 0.0\nsamples = 1\nvalue = 64926.08'),
Text(0.08, 0.2777777777777778, 'poisson = 0.0\nsamples = 1\nvalue = 65200.33'),
Text(0.14, 0.3888888888888889, 'x[1] <= 111785.965\npoisson = 5.355\nsamples = 2\nvalue = 70628.735'),
Text(0.12, 0.2777777777777778, 'poisson = 0.0\nsamples = 1\nvalue = 71498.49'),
Text(0.16, 0.2777777777777778, 'poisson = 0.0\nsamples = 1\nvalue = 69758.98'),
Text(0.14, 0.5, 'poisson = 0.0\nsamples = 1\nvalue = 49490.75'),
Text(0.24, 0.6111111111111112, 'x[2] <= 186014.359\npoisson = 175.115\nsamples = 4\nvalue = 84975.75'),
Text(0.2, 0.5, 'x[2] <= 173897.484\npoisson = 12.01\nsamples = 2\nvalue = 79622.835')]
```

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```
[123]: plt.show()
[124]: y_pred = regressor.predict(x_test)
[125]: #Evaluation Metrics
[126]: from sklearn.metrics import r2_score
[127]: r_score = r2_score(y_test, y_pred)

[128]: r_score
[128]: 0.918880762356684
```


poisson

auto

random

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```
[22]: #Model Creation code for DecisionTree
[23]: from sklearn.tree import DecisionTreeRegressor
[129]: regressor = DecisionTreeRegressor(criterion='poisson', splitter='random', max_features=None)
[130]: regressor = regressor.fit(x_train, y_train)
[131]: #Code to see how the tree is plotted
[132]: import matplotlib.pyplot as plt
[133]: from sklearn import tree
[134]: tree.plot_tree(regressor)
[134]: [Text(0.5018203883495146, 0.9375, 'x[0] <= 67635.975\npoisson = 9248.617\nsamples = 35\nvalue = 110225.321'),
Text(0.2220873786407767, 0.8125, 'x[0] <= 9722.984\npoisson = 5354.266\nsamples = 17\nvalue = 74600.834'),
Text(0.3619538834951457, 0.875, 'True '),
Text(0.0970873786407767, 0.6875, 'x[0] <= 787.599\npoisson = 3672.427\nsamples = 5\nvalue = 41466.274'),
Text(0.05825242718446602, 0.5625, 'x[2] <= 33899.722\npoisson = 2536.582\nsamples = 3\nvalue = 30971.513'),
Text(0.038834951456310676, 0.4375, 'x[0] <= 415.88\npoisson = 151.735\nsamples = 2\nvalue = 39116.57'),
Text(0.019417475728155338, 0.3125, 'poisson = 0.0\nsamples = 1\nvalue = 42559.73'),
Text(0.05825242718446602, 0.3125, 'poisson = -0.0\nsamples = 1\nvalue = 35673.41'),
Text(0.0776990291262135, 0.4375, 'poisson = 0.0\nsamples = 1\nvalue = 14681.4'),
Text(0.13592233009708737, 0.5625, 'x[3] <= 0.113\npoisson = 522.164\nsamples = 2\nvalue = 57208.415'),
Text(0.1165048543693204, 0.4375, 'poisson = 0.0\nsamples = 1\nvalue = 64926.00'),
Text(0.1553390858252427, 0.4375, 'poisson = 0.0\nsamples = 1\nvalue = 49490.75'),
Text(0.3470873786407767, 0.6875, 'x[0] <= 65264.695\npoisson = 1190.527\nsamples = 12\nvalue = 88405.767'),
Text(0.2766990291262136, 0.5625, 'x[0] <= 57193.683\npoisson = 1063.132\nsamples = 10\nvalue = 85113.057'),
Text(0.1941747572815534, 0.4375, 'x[0] <= 27461.987\npoisson = 536.527\nsamples = 7\nvalue = 78051.543'),
Text(0.14563106796116504, 0.3125, 'x[1] <= 133792.383\npoisson = 51.581\nsamples = 3\nvalue = 68819.267'),
Text(0.1262135922330097, 0.1875, 'x[1] <= 99797.354\npoisson = 5.355\nsamples = 2\nvalue = 70628.735'),
Text(0.10679611650485436, 0.0625, 'poisson = 0.0\nsamples = 1\nvalue = 71498.49'),
Text(0.14563106796116504, 0.0625, 'poisson = 0.0\nsamples = 1\nvalue = 6378.401')]
```

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```
Text(0.941747572815534, 0.3125, 'poisson = 0.0\nsamples = 1\nvalue = 191792.06'),
Text(0.9805825242718447, 0.3125, 'poisson = -0.0\nsamples = 1\nvalue = 192261.83')]
```

```
[135]: plt.show()
[136]: y_pred = regressor.predict(x_test)
[137]: #Evaluation Metrics
[138]: from sklearn.metrics import r2_score
[139]: r_score = r2_score(y_test, y_pred)
[140]: r_score
[140]: 0.8876629502172538
```

poisson

sqrt

best

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```
[22]: #Model Creation code for DecisionTree
[23]: from sklearn.tree import DecisionTreeRegressor
[141]: regressor = DecisionTreeRegressor(criterion='poisson', splitter='best', max_features='sqrt')
[142]: regressor = regressor.fit(x_train, y_train)
[143]: #Code to see how the tree is plotted
[144]: import matplotlib.pyplot as plt
[145]: from sklearn import tree
[146]: tree.plot_tree(regressor)
[146]: [Text(0.44436813186813184, 0.9444444444444444, 'x[0] <= 53932.215\npoisson = 9248.617\nsamples = 35\nvalue = 110225.321'),
Text(0.0989010989010989, 0.8333333333333334, 'x[1] <= 51513.145\npoisson = 4562.86\nsamples = 12\nvalue = 62807.681'),
Text(0.27163461538461536, 0.8888888888888888, 'True '),
Text(0.07692307692307693, 0.7222222222222222, 'poisson = 0.0\nsamples = 1\nvalue = 89949.14'),
Text(0.12087912087912088, 0.7222222222222222, 'x[0] <= 771.14\npoisson = 4458.979\nsamples = 11\nvalue = 60348.275'),
Text(0.08593406593406594, 0.6111111111111112, 'x[1] <= 126205.359\npoisson = 2536.582\nsamples = 3\nvalue = 30971.513'),
Text(0.04395604395604396, 0.5, 'x[0] <= 271.025\npoisson = 2256.033\nsamples = 2\nvalue = 25177.405'),
Text(0.02197802197802198, 0.3888888888888889, 'poisson = 0.0\nsamples = 1\nvalue = 14681.4'),
Text(0.06593406593406594, 0.3888888888888889, 'poisson = 0.0\nsamples = 1\nvalue = 35673.41'),
Text(0.08791208791208792, 0.5, 'poisson = 0.0\nsamples = 1\nvalue = 42559.73'),
Text(0.17582417582417584, 0.6111111111111112, 'x[0] <= 26152.345\npoisson = 963.594\nsamples = 8\nvalue = 71353.561'),
Text(0.13186813186813187, 0.5, 'x[2] <= 222557.789\npoisson = 495.34\nsamples = 5\nvalue = 64174.926'),
Text(0.10989010989010989, 0.3888888888888889, 'x[0] <= 22909.335\npoisson = 59.835\nsamples = 4\nvalue = 67845.97'),
Text(0.08791208791208792, 0.2777777777777778, 'x[2] <= 31934.446\npoisson = 36.59\nsamples = 3\nvalue = 66628.463'),
Text(0.06593406593406594, 0.16666666666666666, 'x[0] <= 11588.985\npoisson = 0.145\nsamples = 2\nvalue = 65063.205'),
Text(0.04395604395604396, 0.05555555555555555, 'poisson = 0.0\nsamples = 1\nvalue = 64926.08'),
Text(0.08791208791208792, 0.05555555555555555, 'poisson = 0.0\nsamples = 1\nvalue = 65200.33')]
```

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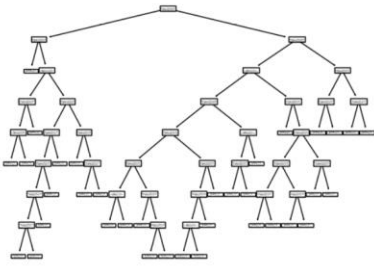
Code Snippets

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```
[147]: plt.show()
[148]: y_pred = regressor.predict(x_test)
[149]: #Evaluation Metrics
[150]: from sklearn.metrics import r2_score
[151]: r_score = r2_score(y_test, y_pred)
[152]: r_score
[152]: 0.42313512471580794
```

poisson

sqrt

random

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```
[23]: from sklearn.tree import DecisionTreeRegressor

[153]: regressor = DecisionTreeRegressor(criterion='poisson', splitter='random', max_features='sqrt')

[154]: regressor = regressor.fit(x_train, y_train)

[155]: #Code to see how the tree is plotted

[156]: import matplotlib.pyplot as plt

[157]: from sklearn import tree

[158]: tree.plot_tree(regressor)

[158]: [Text(0.6753784295175024, 0.9583333333333334, 'x[1] <= 153115.149\npoisson = 9248.617\nnsamples = 35\nnvalue = 110225.321'),
Text(0.4718543046357616, 0.875, 'x[0] <= 6397.094\npoisson = 10002.355\nnsamples = 31\nnvalue = 110515.915'),
Text(0.573616367076632, 0.9166666666666667, 'True '),
Text(0.336802270577105, 0.7916666666666666, 'x[2] <= 170009.203\npoisson = 3672.427\nnsamples = 5\nnvalue = 41466.274'),
Text(0.3065279091769158, 0.7083333333333334, 'x[0] <= 337.907\npoisson = 4358.528\nnsamples = 4\nnvalue = 39460.155'),
Text(0.24597918637653737, 0.625, 'x[1] <= 132582.519\npoisson = 3543.247\nnsamples = 2\nnvalue = 28620.565'),
Text(0.21570482497634816, 0.5416666666666666, 'poisson = 0.0\nnsamples = 1\nnvalue = 14681.4'),
Text(0.27625354777672656, 0.5416666666666666, 'poisson = 0.0\nnsamples = 1\nnvalue = 42559.73'),
Text(0.36707663197729423, 0.625, 'x[1] <= 75045.692\npoisson = 2157.579\nnsamples = 2\nnvalue = 50299.745'),
Text(0.336802270577105, 0.5416666666666666, 'poisson = 0.0\nnsamples = 1\nnvalue = 35673.41'),
Text(0.3973509933774834, 0.5416666666666666, 'poisson = 0.0\nnsamples = 1\nnvalue = 64926.08'),
Text(0.36707663197729423, 0.7083333333333334, 'poisson = 0.0\nnsamples = 1\nnvalue = 49490.75'),
Text(0.6009063386944181, 0.7916666666666666, 'x[2] <= 57373.86\npoisson = 4990.314\nnsamples = 26\nnvalue = 123794.692'),
Text(0.4062819299905326, 0.7083333333333334, 'x[1] <= 127500.31\npoisson = 1150.06\nnsamples = 2\nnvalue = 83593.41'),
Text(0.45600756859035, 0.625, 'poisson = 0.0\nnsamples = 1\nnvalue = 69758.98'),
Text(0.5165562913987285, 0.625, 'poisson = -0.0\nnsamples = 1\nnvalue = 97427.84'),
Text(0.727530747398297, 0.7083333333333334, 'x[0] <= 124306.569\npoisson = 4650.616\nnsamples = 24\nnvalue = 127144.799'),
Text(0.577105041911069, 0.625, 'x[0] <= 119313.808\npoisson = 2554.393\nnsamples = 18\nnvalue = 111980.726'),
Text(0.45709971617780185, 0.5416666666666666, 'x[0] <= 84490.463\npoisson = 2080.553\nnsamples = 16\nnvalue = 107105.083'),
Text(0.31031220435193946, 0.4583333333333333, 'x[0] <= 66607.681\npoisson = 1317.372\nnsamples = 12\nnvalue = 98541.754'),
Text(0.1665089877010407, 0.375, 'x[0] <= 34856.221\npoisson = 759.954\nnsamples = 8\nnvalue = 89968.508'),
Text(0.06054872280037843, 0.2916666666666667, 'x[1] <= 107460.083\npoisson = 391.463\nnsamples = 3\nnvalue = 80148.863'),
Text(0.030374361400180316, 0.3083333333333333, 'poisson = 0.0\nnsamples = 1\nnvalue = 71400.40')]
```

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```
[159]: plt.show()

[160]: y_pred = regressor.predict(x_test)

[161]: #Evaluation Metrics

[162]: from sklearn.metrics import r2_score

[163]: r_score = r2_score(y_test, y_pred)

[164]: r_score

[164]: 0.7524659638413829
```

poisson

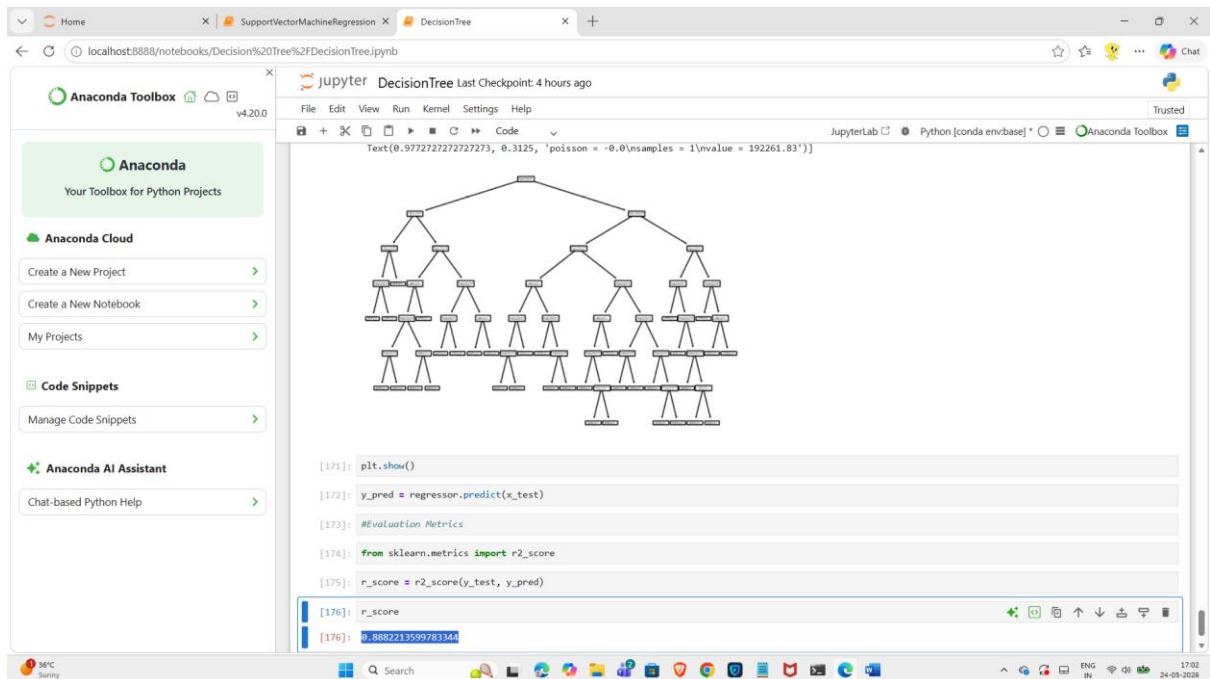
Log2

best

```
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JupyterLab Python [conda envbase] Anaconda Toolbox

[22]: #Model Creation code for DecisionTree
[23]: from sklearn.tree import DecisionTreeRegressor
[165]: regressor = DecisionTreeRegressor(criterion='poisson', splitter='best', max_features='log2')
[166]: regressor = regressor.fit(x_train, y_train)
[167]: #Code to see how the tree is plotted
[168]: import matplotlib.pyplot as plt
[169]: from sklearn import tree
[170]: tree.plot_tree(regressor)

[170]: [Text(0.43323863636363635, 0.9375, 'x[0] <= 53032.215\npoisson = 9248.617\nsamples = 35\nvalue = 110225.321'),
Text(0.13636363636363635, 0.8125, 'x[0] <= 771.14\npoisson = 4562.86\nsamples = 12\nvalue = 62807.681'),
Text(0.28489113636363635, 0.875, 'True'),
Text(0.06818181818181818, 0.6875, 'x[0] <= 271.025\npoisson = 2536.582\nsamples = 3\nvalue = 38971.513'),
Text(0.045454545454545456, 0.5625, 'x[1] <= 126205.359\npoisson = 3543.247\nsamples = 2\nvalue = 28620.565'),
Text(0.022727272727272728, 0.4375, 'poisson = 0.0\nsamples = 1\nvalue = 14681.4'),
Text(0.06818181818181818, 0.4375, 'poisson = 0.0\nsamples = 1\nvalue = 42559.73'),
Text(0.09090909090909091, 0.5625, 'poisson = 0.0\nsamples = 1\nvalue = 35673.41'),
Text(0.20454545454545456, 0.6875, 'x[0] <= 26152.345\npoisson = 1875.385\nsamples = 9\nvalue = 73419.737'),
Text(0.13636363636363635, 0.5625, 'x[3] <= 0.5\npoisson = 495.34\nsamples = 5\nvalue = 64174.926'),
Text(0.11363636363636363, 0.4375, 'x[2] <= 31934.446\npoisson = 59.835\nsamples = 4\nvalue = 67845.97'),
Text(0.06818181818181818, 0.3125, 'x[0] <= 11588.985\npoisson = 0.145\nsamples = 2\nvalue = 65063.205'),
Text(0.045454545454545456, 0.1875, 'poisson = 0.0\nsamples = 1\nvalue = 64926.08'),
Text(0.09090909090909091, 0.1875, 'poisson = 0.0\nsamples = 1\nvalue = 65200.33'),
Text(0.15909090909090909, 0.3125, 'x[4] <= 0.5\npoisson = 5.355\nsamples = 2\nvalue = 70628.735'),
Text(0.13636363636363635, 0.1875, 'poisson = 0.0\nsamples = 1\nvalue = 71498.49'),
Text(0.18181818181818182, 0.1875, 'poisson = 0.0\nsamples = 1\nvalue = 69758.98'),
Text(0.15909090909090909, 0.4375, 'poisson = 0.0\nsamples = 1\nvalue = 48490.75'),
Text(0.2727272727272727, 0.5625, 'x[2] <= 186014.359\npoisson = 175.115\nsamples = 4\nvalue = 84975.75'),
Text(0.22727272727272727, 0.4375, 'x[0] <= 33656.421\npoisson = 12.01\nsamples = 2\nvalue = 79622.835'),
Text(0.20454545454545456, 0.3125, 'poisson = 0.0\nsamples = 1\nvalue = 78239.91'),
Text(0.36, 0.3125, 'poisson = 0.0\nsamples = 1\nvalue = 81000.36')]
```



poisson

Log2

random

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```
[22]: #Model Creation code for DecisionTree
[23]: from sklearn.tree import DecisionTreeRegressor

[177]: regressor = DecisionTreeRegressor(criterion='poisson', splitter='random', max_features='log2')
[178]: regressor = regressor.fit(x_train, y_train)

[179]: #Code to see how the tree is plotted
[180]: import matplotlib.pyplot as plt
[181]: from sklearn import tree
[182]: tree.plot_tree(regressor)

[182]: [Text(0.6291118421052632, 0.9583333333333334, 'x[0] <= 131372.174\npoisson = 9248.617\nnsamples = 35\nnvalue = 110225.321'),
Text(0.38980263157894735, 0.875, 'x[1] <= 82746.576\npoisson = 7334.133\nnsamples = 30\nnvalue = 99260.557'),
Text(0.5094572368421053, 0.9166666666666667, 'True '),
Text(0.2565789473684211, 0.7916666666666666, 'x[2] <= 178101.707\npoisson = 6060.060\nnsamples = 2\nnvalue = 62811.275'),
Text(0.23026315789473684, 0.7083333333333334, 'poisson = 0.0\nnsamples = 1\nnvalue = 35673.41'),
Text(0.28289473684210525, 0.7083333333333334, 'poisson = 0.0\nnsamples = 1\nnvalue = 89949.14'),
Text(0.5230263157894737, 0.7916666666666666, 'x[0] <= 95539.544\npoisson = 6840.86\nnsamples = 28\nnvalue = 101864.077'),
Text(0.3355263157894737, 0.7083333333333334, 'x[0] <= 31869.513\npoisson = 6135.424\nnsamples = 23\nnvalue = 92693.484'),
Text(0.23684210526315789, 0.625, 'x[2] <= 218950.62\npoisson = 4436.063\nnsamples = 9\nnvalue = 60784.873'),
Text(0.21052631578947367, 0.5416666666666666, 'x[4] <= 0.977\npoisson = 4834.171\nnsamples = 8\nnvalue = 62196.639'),
Text(0.15789473684210525, 0.4583333333333333, 'x[3] <= 0.096\npoisson = 6338.155\nnsamples = 6\nnvalue = 60481.342'),
Text(0.13157894736842105, 0.375, 'x[1] <= 126797.896\npoisson = 5985.387\nnsamples = 5\nnvalue = 54435.972'),
Text(0.07894736842105263, 0.2916666666666667, 'x[0] <= 24468.53\npoisson = 9173.743\nnsamples = 3\nnvalue = 54806.6'),
Text(0.05263157894736842, 0.20833333333333334, 'x[0] <= 16374.373\npoisson = 10199.148\nnsamples = 2\nnvalue = 43089.945'),
Text(0.02631578947368421, 0.125, 'poisson = 0.0\nnsamples = 1\nnvalue = 16681.4'),
Text(0.07894736842105263, 0.125, 'poisson = 0.0\nnsamples = 1\nnvalue = 71498.49'),
Text(0.10526315789473684, 0.20833333333333334, 'poisson = 0.0\nnsamples = 1\nnvalue = 78239.91'),
Text(0.18471063615789473, 0.7083333333333334, 'x[3] <= 77733.434\npoisson = 5100.516\nnsamples = 3\nnvalue = 63000.03')]
```

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```
[183]: plt.show()
[184]: y_pred = regressor.predict(x_test)

[185]: #Evaluation Metrics
[186]: from sklearn.metrics import r2_score
[187]: r_score = r2_score(y_test, y_pred)

[188]: r_score
[188]: 0.945481825284668
```

Friedman_mse

auto

best

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```
[22]: #MODEL CREATION CODE FOR DECISIONTREE

[23]: from sklearn.tree import DecisionTreeRegressor

[189]: regressor = DecisionTreeRegressor(criterion='friedman_mse', splitter='best', max_features=None)

[190]: regressor = regressor.fit(x_train, y_train)

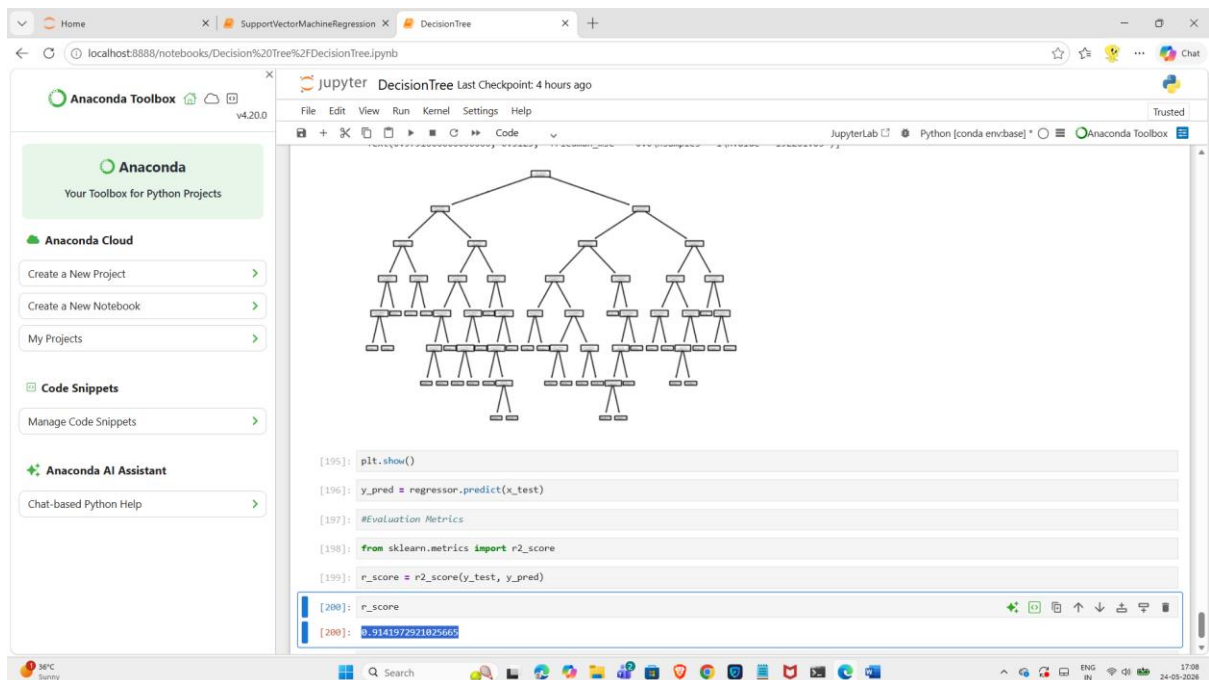
[191]: #Code to see how the tree is plotted

[192]: import matplotlib.pyplot as plt

[193]: from sklearn import tree

[194]: tree.plot_tree(regressor)

[194]: [Text(0.47265625, 0.9375, 'x[0] <= 66569.004\nfriedman_mse = 185097680.97\nnsamples = 35\nnvalue = 110225.321'),
Text(0.203125, 0.8125, 'x[0] <= 8410.595\nfriedman_mse = 649408291.256\nnsamples = 16\nnvalue = 72466.661'),
Text(0.337890625, 0.875, 'True '),
Text(0.10416666666666667, 0.6875, 'x[0] <= 771.14\nfriedman_mse = 273387421.435\nnsamples = 5\nnvalue = 41466.274'),
Text(0.0625, 0.5625, 'x[2] <= 22586.529\nfriedman_mse = 140587463.397\nnsamples = 3\nnvalue = 30971.513'),
Text(0.041666666666666664, 0.4375, 'x[1] <= 93585.035\nfriedman_mse = 11855350.786\nnsamples = 2\nnvalue = 39116.57'),
Text(0.020833333333333332, 0.3125, 'friedman_mse = 0.0\nnsamples = 1\nnvalue = 35673.41'),
Text(0.0625, 0.3125, 'friedman_mse = -0.0\nnsamples = 1\nnvalue = 42559.73'),
Text(0.08333333333333333, 0.4375, 'friedman_mse = 0.0\nnsamples = 1\nnvalue = 14681.4'),
Text(0.14583333333333334, 0.5625, 'x[1] <= 119984.625\nfriedman_mse = 59562353.052\nnsamples = 2\nnvalue = 57208.415'),
Text(0.125, 0.4375, 'friedman_mse = 0.0\nnsamples = 1\nnvalue = 49490.75'),
Text(0.16666666666666666, 0.4375, 'friedman_mse = 0.0\nnsamples = 1\nnvalue = 64926.00'),
Text(0.30208333333333333, 0.6875, 'x[0] <= 41314.23\nfriedman_mse = 184939093.473\nnsamples = 11\nnvalue = 86557.746'),
Text(0.25, 0.5625, 'x[0] <= 26152.345\nfriedman_mse = 70370705.683\nnsamples = 6\nnvalue = 76068.61'),
Text(0.20833333333333334, 0.4375, 'x[1] <= 141094.219\nfriedman_mse = 7052667.139\nnsamples = 3\nnvalue = 68819.267'),
Text(0.1875, 0.3125, 'x[0] <= 19573.33\nfriedman_mse = 756473.76\nnsamples = 2\nnvalue = 70628.735'),
Text(0.16666666666666666, 0.1875, 'friedman_mse = 0.0\nnsamples = 1\nnvalue = 69758.90'),
Text(0.20833333333333334, 0.1875, 'friedman_mse = 0.0\nnsamples = 1\nnvalue = 71498.49'),
Text(0.22916666666666666, 0.3125, 'friedman_mse = -0.0\nnsamples = 1\nnvalue = 65200.33'),
Text(0.29166666666666667, 0.4375, 'x[2] <= 180863.055\nfriedman_mse = 20582786.690\nnsamples = 3\nnvalue = 83317.953'),
Text(0.27083333333333333, 0.3125, 'x[1] <= 100764.07\nfriedman_mse = 1912481.556\nnsamples = 2\nnvalue = 79622.835'),
Text(0.25, 0.1875, 'friedman_mse = 0.0\nnsamples = 1\nnvalue = 81005.76'),
```



Friedman_mse

auto

random

```
from sklearn.tree import DecisionTreeRegressor

regressor = DecisionTreeRegressor(criterion='friedman_mse', splitter='random', max_features=None)

regressor = regressor.fit(x_train, y_train)

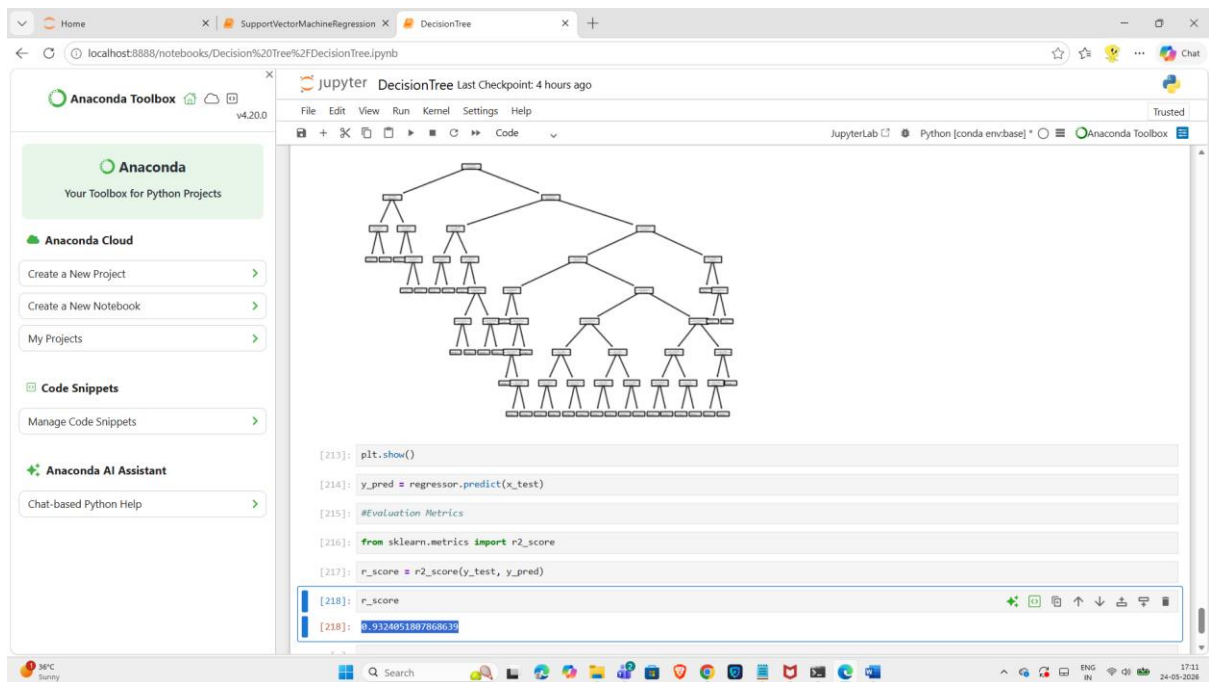
#Code to see how the tree is plotted

import matplotlib.pyplot as plt

from sklearn import tree

tree.plot_tree(regressor)

[208]: [Text(0.28744183773584906, 0.9444444444444444, 'x[0] <= 2900.127\nfriedman_mse = 1850097680.97\nnsamples = 35\nnvalue = 110225.321'),
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Text(0.1814563679245283, 0.8888888888888888, 'True'),
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Text(0.11320754716981132, 0.7222222222222222, 'friedman_mse = 0.0\nnsamples = 1\nnvalue = 14681.4'),
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Text(0.1320754716981132, 0.6111111111111112, 'x[2] <= 69978.971\nfriedman_mse = 59562353.052\nnsamples = 2\nnvalue = 57208.415'),
Text(0.11320754716981132, 0.5, 'friedman_mse = 0.0\nnsamples = 1\nnvalue = 64926.08'),
Text(0.1509433962264151, 0.5, 'friedman_mse = 0.0\nnsamples = 1\nnvalue = 49490.75'),
Text(0.4994103773584906, 0.8333333333333334, 'x[0] <= 45672.223\nfriedman_mse = 1193587018.73\nnsamples = 30\nnvalue = 121685.162'),
Text(0.3934257075471698, 0.8888888888888888, 'False'),
Text(0.24528301886792453, 0.7222222222222222, 'x[2] <= 39044.385\nfriedman_mse = 83909884.024\nnsamples = 7\nnvalue = 78051.543'),
Text(0.20754716981132076, 0.6111111111111112, 'x[1] <= 143652.828\nfriedman_mse = 5195322.456\nnsamples = 2\nnvalue = 67479.655'),
Text(0.18867924528301888, 0.5, 'friedman_mse = 0.0\nnsamples = 1\nnvalue = 69758.98'),
Text(0.22641509433962265, 0.5, 'friedman_mse = -0.0\nnsamples = 1\nnvalue = 65200.33'),
Text(0.2830188679245283, 0.6111111111111112, 'x[2] <= 158610.328\nfriedman_mse = 52807413.447\nnsamples = 5\nnvalue = 82280.298'),
Text(0.2641509433962264, 0.5, 'friedman_mse = 0.0\nnsamples = 1\nnvalue = 71498.49'),
Text(0.3018867924528302, 0.5, 'x[2] <= 183833.256\nfriedman_mse = 29681959.388\nnsamples = 4\nnvalue = 84975.75'),
Text(0.2641509433962264, 0.8888888888888889, 'x[1] <= 118384.561\nfriedman_mse = 1912481.556\nnsamples = 2\nnvalue = 79622.835'),
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Text(0.3830188679245283, 0.3777777777777778, 'friedman_mse = 0.0\nnsamples = 1\nnvalue = 79330.051')]
```



Friedman_mse

sqrt

best

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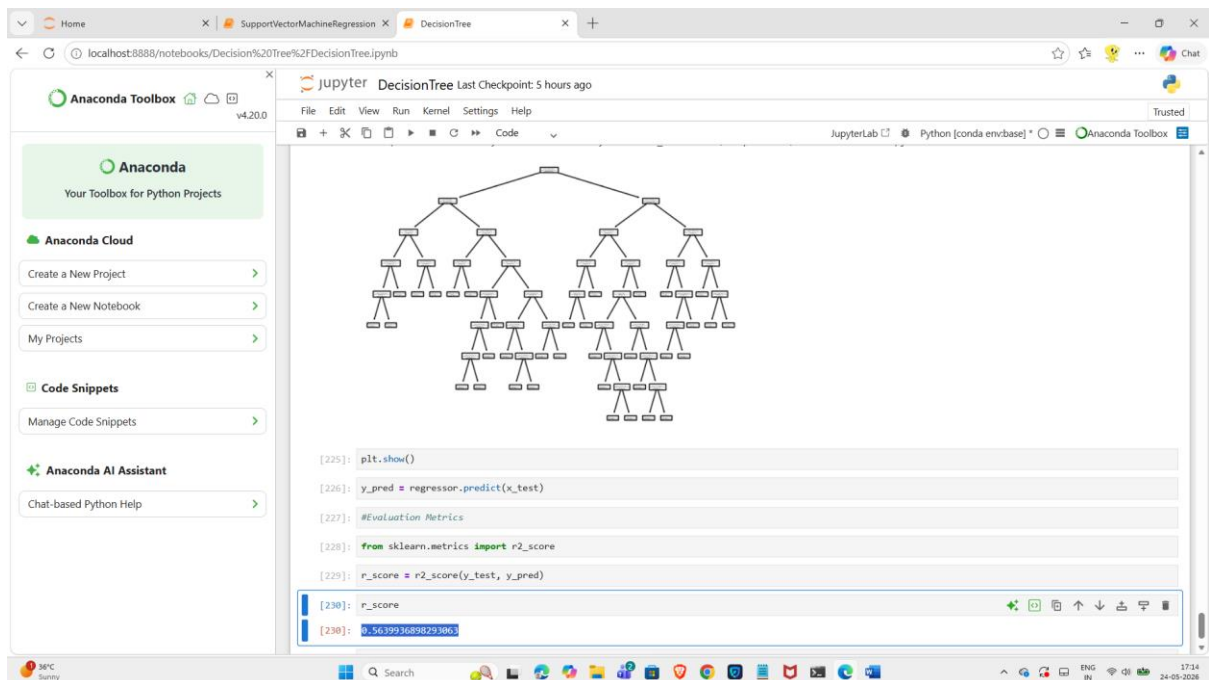
Anaconda AI Assistant

Chat-based Python Help

JupyterLab Python [conda envbase] Trusted

```
[221]: #Model Creation code for DecisionTree
[222]:
[223]: from sklearn.tree import DecisionTreeRegressor
[224]:
[219]: regressor = DecisionTreeRegressor(criterion='friedman_mse', splitter='best', max_features='sqrt')
[220]: regressor = regressor.fit(x_train, y_train)
[221]: #Code to see how the tree is plotted
[222]:
[223]: import matplotlib.pyplot as plt
[224]:
[225]: from sklearn import tree
[226]:
[227]: tree.plot_tree(regressor)
[228]:
```

[Text(0.4955357142857143, 0.9444444444444444, 'x[0] <= 66569.004\nfriedman_mse = 1850097680.97\nnsamples = 35\nvalue = 110225.321'), Text(0.22321428571428573, 0.8333333333333334, 'x[0] <= 8410.595\nfriedman_mse = 649408291.256\nnsamples = 16\nvalue = 72466.661'), Text(0.359375, 0.8888888888888888, 'True '), Text(0.11904761904761904, 0.7222222222222222, 'x[0] <= 771.14\nfriedman_mse = 273387421.435\nnsamples = 5\nvalue = 41466.274'), Text(0.07142857142857142, 0.6111111111111112, 'x[1] <= 126205.350\nfriedman_mse = 140587463.397\nnsamples = 3\nvalue = 30971.513'), Text(0.047619047619047616, 0.5, 'x[2] <= 22586.529\nfriedman_mse = 110166120.96\nnsamples = 2\nvalue = 25177.405'), Text(0.023809523809523808, 0.3888888888888889, 'friedman_mse = 0.0\nnsamples = 1\nvalue = 35673.41'), Text(0.07142857142857142, 0.3888888888888889, 'friedman_mse = 0.0\nnsamples = 1\nvalue = 14681.4'), Text(0.09523809523809523, 0.5, 'friedman_mse = -0.0\nnsamples = 1\nvalue = 42559.73'), Text(0.16666666666666666, 0.6111111111111112, 'x[1] <= 119984.025\nfriedman_mse = 59563353.052\nnsamples = 2\nvalue = 57208.415'), Text(0.14285714285714285, 0.5, 'friedman_mse = 0.0\nnsamples = 1\nvalue = 49498.75'), Text(0.19047619047619047, 0.5, 'friedman_mse = 0.0\nnsamples = 1\nvalue = 64926.08'), Text(0.3273809523809524, 0.7222222222222222, 'x[2] <= 48009.711\nfriedman_mse = 184930093.473\nnsamples = 11\nvalue = 86557.746'), Text(0.2619047619047619, 0.6111111111111112, 'x[0] <= 18841.735\nfriedman_mse = 5195322.456\nnsamples = 2\nvalue = 67479.655'), Text(0.23809523809523808, 0.5, 'friedman_mse = 0.0\nnsamples = 1\nvalue = 60758.98'), Text(0.2857142857142857, 0.5, 'friedman_mse = -0.0\nnsamples = 1\nvalue = 65200.33'), Text(0.39285714285714285, 0.6111111111111112, 'x[2] <= 142981.867\nfriedman_mse = 126025134.665\nnsamples = 9\nvalue = 90797.322'), Text(0.3333333333333333, 0.5, 'x[2] <= 122550.504\nfriedman_mse = 13529375.432\nnsamples = 4\nvalue = 101443.603'), Text(0.30952380952380953, 0.3888888888888889, 'x[0] <= 64507.168\nfriedman_mse = 2247892.112\nnsamples = 3\nvalue = 59456.69'), Text(0.2857142857142857, 0.2777777777777778, 'x[1] <= 122430.445\nfriedman_mse = 1574711.266\nnsamples = 3\nvalue = 98682.715'), Text(0.2619047619047619, 0.16666666666666666, 'friedman_mse = 0.0\nnsamples = 1\nvalue = 99937.59'), Text(0.30952380952380953, 0.16666666666666666, 'friedman_mse = 0.0\nnsamples = 1\nvalue = 97427.84'),



Friedman_mse

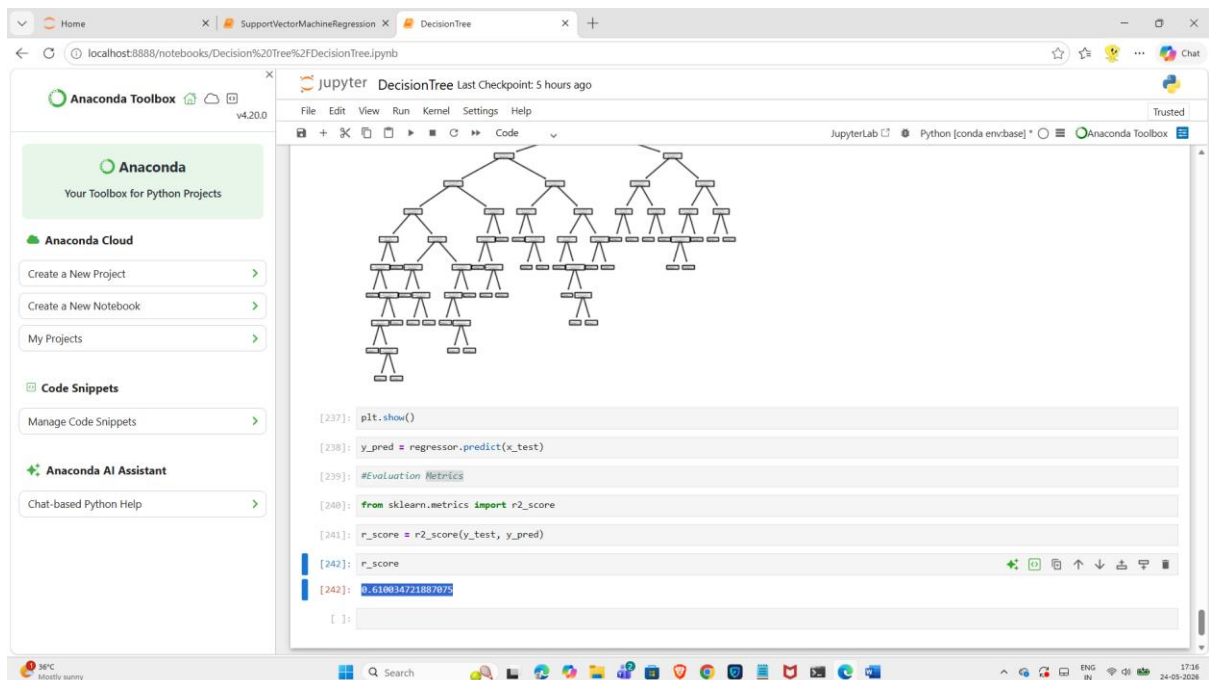
sqrt

random

```
[22]: #Model Creation code for DecisionTree
[23]: from sklearn.tree import DecisionTreeRegressor

[231]: regressor = DecisionTreeRegressor(criterion='friedman_mse', splitter='random', max_features='sqrt')
[232]: regressor = regressor.fit(x_train, y_train)
[233]: #Code to see how the tree is plotted
[234]: import matplotlib.pyplot as plt
[235]: from sklearn import tree
[236]: tree.plot_tree(regressor)

[236]: [Text(0.6085434782608695, 0.95, 'x[2] <= 299224.109\nfriedman_mse = 1850097680.97\nsamples = 35\nvalue = 110225.321'),
Text(0.375, 0.85, 'x[1] <= 130599.462\nfriedman_mse = 1232451653.682\nsamples = 26\nvalue = 94467.995'),
Text(0.48777173913043476, 0.8999999999999999, 'True '),
Text(0.2391304347826087, 0.75, 'x[0] <= 76413.834\nfriedman_mse = 1092781355.216\nsamples = 17\nvalue = 86401.124'),
Text(0.13043478260869565, 0.65, 'x[2] <= 115328.916\nfriedman_mse = 730389288.622\nsamples = 13\nvalue = 73982.428'),
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Text(0.021739130434782608, 0.35, 'friedman_mse = 0.0\nsamples = 1\nvalue = 35673.41'),
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Text(0.021739130434782608, 0.15, 'friedman_mse = 0.0\nsamples = 1\nvalue = 14681.4'),
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Text(0.08695652173913043, 0.05, 'friedman_mse = -0.0\nsamples = 1\nvalue = 69758.98'),
Text(0.08695652173913043, 0.25, 'friedman_mse = -0.0\nsamples = 1\nvalue = 97427.84'),
Text(0.08695652173913043, 0.45, 'friedman_mse = 0.0\nsamples = 1\nvalue = 99937.59'),
Text(0.1956521739130435, 0.55, 'x[0] <= 28724.887\nfriedman_mse = 378217466.549\nsamples = 7\nvalue = 82766.61'),
Text(0.13043478260869565, 0.45, 'x[0] <= 15058.614\nfriedman_mse = 283581172.683\nsamples = 3\nvalue = 70565.81'),
Text(0.10869565217391304, 0.35, 'friedman_mse = 0.0\nsamples = 1\nvalue = 49490.75'),
Text(0.1517391304347827, 0.35, 'x[0] <= 75430.563\nfriedman_mse = 93793181.633\nsamples = 2\nvalue = 81103.34')]
```



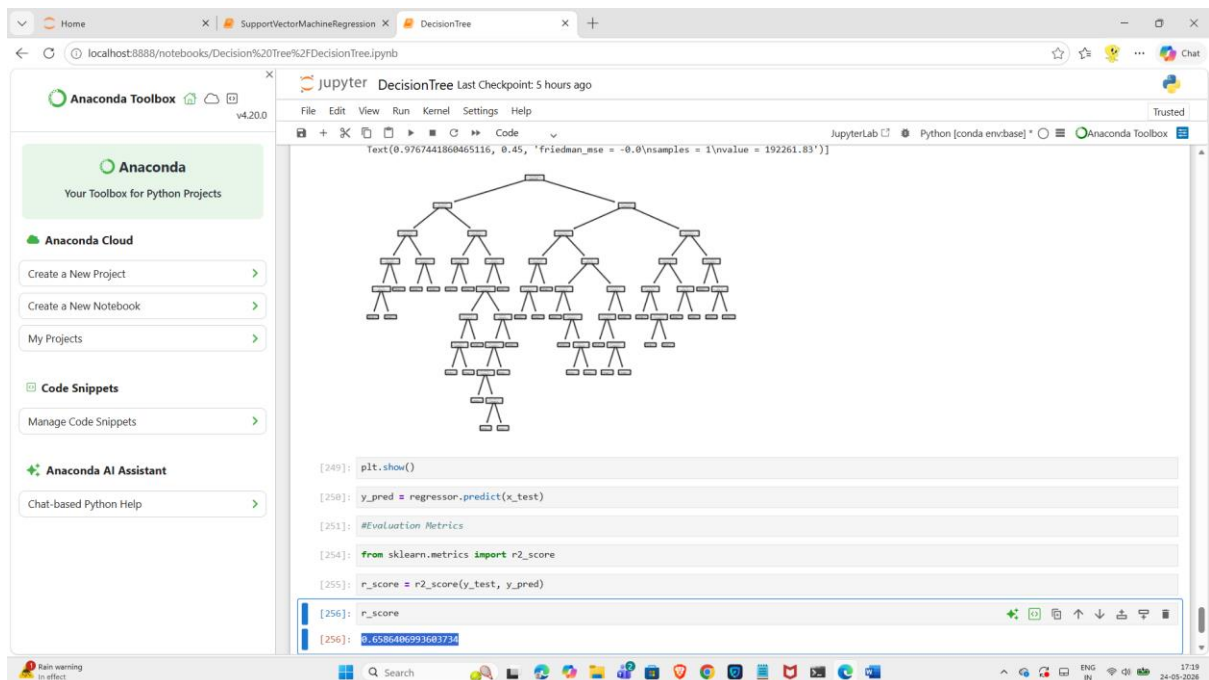
Friedman_mse

Log2

best

DecisionTree Last Checkpoint: 5 hours ago

```
[22]: #Model Creation code for DecisionTree
[23]: from sklearn.tree import DecisionTreeRegressor
[24]: regressor = DecisionTreeRegressor(criterion='friedman_mse', splitter='best', max_features='log2')
[244]: regressor = regressor.fit(x_train, y_train)
[245]: #Code to see how the tree is plotted
[246]: import matplotlib.pyplot as plt
[247]: from sklearn import tree
[248]: tree.plot_tree(regressor)
[248]: [Text(0.4563953488372093, 0.95, 'x[0] <= 66569.004\nfriedman_mse = 1850097680.97\nnsamples = 35\nnvalue = 110225.321'),
Text(0.20930232558139536, 0.85, 'x[0] <= 8410.595\nfriedman_mse = 649408291.256\nnsamples = 16\nnvalue = 72466.661'),
Text(0.3328488372093023, 0.8999999999999999, 'True '),
Text(0.11627906976744186, 0.75, 'x[2] <= 23538.494\nfriedman_mse = 273387421.435\nnsamples = 5\nnvalue = 41466.274'),
Text(0.06976744186046512, 0.65, 'x[0] <= 771.14\nfriedman_mse = 155932635.288\nnsamples = 3\nnvalue = 47719.74'),
Text(0.046511627906976744, 0.55, 'x[4] <= 0.5\nfriedman_mse = 11895350.786\nnsamples = 2\nnvalue = 39116.57'),
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Text(0.06976744186046512, 0.45, 'friedman_mse = -0.0\nnsamples = 1\nnvalue = 35673.41'),
Text(0.09302325581395349, 0.55, 'friedman_mse = 0.0\nnsamples = 1\nnvalue = 64926.08'),
Text(0.16279069767441862, 0.65, 'x[2] <= 171143.764\nfriedman_mse = 302922711.856\nnsamples = 2\nnvalue = 32086.075'),
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Text(0.1804651162790697, 0.55, 'friedman_mse = 0.0\nnsamples = 1\nnvalue = 69490.75'),
Text(0.3023255813953488, 0.75, 'x[2] <= 40809.711\nfriedman_mse = 184939093.473\nnsamples = 11\nnvalue = 86557.746'),
Text(0.2558139534883721, 0.65, 'x[0] <= 18841.735\nfriedman_mse = 5195322.456\nnsamples = 2\nnvalue = 67479.655'),
Text(0.23255813953488372, 0.55, 'friedman_mse = 0.0\nnsamples = 1\nnvalue = 69758.98'),
Text(0.13953488372093023, 0.55, 'friedman_mse = -0.0\nnsamples = 1\nnvalue = 65200.33'),
Text(0.3488372093023256, 0.65, 'x[4] <= 0.5\nfriedman_mse = 126025134.665\nnsamples = 9\nnvalue = 90797.322'),
Text(0.32558139534883723, 0.55, 'x[2] <= 142981.867\nfriedman_mse = 127126651.093\nnsamples = 8\nnvalue = 89521.408'),
Text(0.27906976744186046, 0.45, 'x[2] <= 114546.934\nfriedman_mse = 17953528.097\nnsamples = 3\nnvalue = 101589.923'),
Text(0.2558139534883721, 0.35, 'x[2] <= 68608.246\nfriedman_mse = 1574711.266\nnsamples = 2\nnvalue = 98682.715'),
Text(0.23255813953488372, 0.25, 'friedman_mse = 0.0\nnsamples = 1\nnvalue = 97427.84'),
Text(0.07606076744186046, 0.15, 'friedman_mse = 0.0\nnsamples = 1\nnvalue = 60037.60')]
```



Friedman_mse

Log2

random

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```
[22]: #Model Creation code for DecisionTree
[23]: from sklearn.tree import DecisionTreeRegressor
[257]: regressor = DecisionTreeRegressor(criterion='friedman_mse', splitter='random', max_features='log2')
[258]: regressor = regressor.fit(x_train, y_train)
[259]: #Code to see how the tree is plotted
[260]: import matplotlib.pyplot as plt
[261]: from sklearn import tree
[262]: tree.plot_tree(regressor)
[262]: [Text(0.49166666666666664, 0.9545454545454546, 'x[2] <= 207780.41\nfriedman_mse = 1850097680.97\nsamples = 35\nvalue = 110225.321'),
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Text(0.17777777777777778, 0.6818181818181818, 'x[0] <= 54304.645\nfriedman_mse = 767078558.909\nsamples = 12\nvalue = 71806.427'),
Text(0.13333333333333333, 0.5909090909090909, 'x[1] <= 135007.986\nfriedman_mse = 540520737.109\nsamples = 9\nvalue = 60921.433'),
Text(0.11111111111111111, 0.5, 'x[4] <= 0.362\nfriedman_mse = 500673808.252\nsamples = 8\nvalue = 63216.646'),
Text(0.04444444444444444, 0.4090909090909091, 'x[0] <= 11094.065\nfriedman_mse = 721320830.989\nsamples = 5\nvalue = 67074.94'),
Text(0.02222222222222222, 0.3181818181818182, 'friedman_mse = 0.0\nsamples = 1\nvalue = 14681.4'),
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Text(0.04444444444444444, 0.22727272727272727, 'friedman_mse = 0.0\nsamples = 1\nvalue = 71498.49'),
Text(0.08888888888888889, 0.22727272727272727, 'x[2] <= 183428.177\nfriedman_mse = 24971115.471\nsamples = 3\nvalue = 83064.937'),
Text(0.06666666666666667, 0.13636363636363635, 'x[2] <= 172919.696\nfriedman_mse = 1912481.556\nsamples = 2\nvalue = 79622.835'),
Text(0.04444444444444444, 0.04545454545454546, 'friedman_mse = 0.0\nsamples = 1\nvalue = 78239.91'),
Text(0.08888888888888889, 0.04545454545454546, 'friedman_mse = 0.0\nsamples = 1\nvalue = 81005.76'),
Text(0.11111111111111111, 0.13636363636363635, 'friedman_mse = -0.0\nsamples = 1\nvalue = 89949.14'),
Text(0.17777777777777778, 0.4090909090909091, 'x[2] <= 15999.147\nfriedman_mse = 226766856.307\nsamples = 3\nvalue = 56786.157'),
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```

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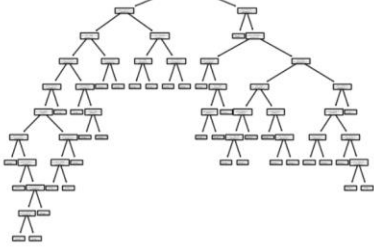
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```
[263]: plt.show()
[264]: y_pred = regressor.predict(x_test)
[265]: #Evaluation Metrics
[266]: from sklearn.metrics import r2_score
[267]: r_score = r2_score(y_test, y_pred)
[268]: r_score
[268]: 0.25801938600151377
[ ]:
```